FLAIR Task Schematic

ICU Daily LTACH Discharge at 07:00 Local (Task T2)

FLAIR Benchmark

June 4, 2026

Abstract

We specify the FLAIR daily ICU LTACH-discharge task: at every local 07:00 rounds anchor during the first ICU stay, predict whether the hospitalization will end with transfer to a Long-Term Acute Care Hospital (LTACH) rather than in-hospital mortality (Expired or Hospice). Task T2 is the exact label complement of Task T1 on the identical cohort, identical grid, and identical 24-hour lookback. This schematic specifies only the differences; the shared pipeline is the authority defined in `icu_daily_mortality.tex` (Task T1).

1 Relationship to Task T1

The build pipeline of Task T2 is byte-identical to Task T1's: same cohort filters (adult, ≥ 1 ICU ADT, `discharge_category` $\in$ {LTACH, Expired, Hospice}), same pred_start, pred_stop, and decision-point grid, same 24h lookback, same temporal split. The only difference is the final label rule. The full operational definition, cohort details, and step-by-step build logic are in `icu_daily_mortality.tex` (Task T1); the flowchart of Figure 1 is shared.

2 Label rule (Task T2 only)

Let $\texttt{disp}_i = \texttt{lower}(\texttt{strip}(\texttt{discharge_category}_i))$. For d in the daily 07:00 decision grid D_i:

$$y_i(d) \;=\; \mathbb{1}[\,\texttt{disp}_i \text{ matches regex } \texttt{ltach|long.?term}\,]\,.$$

- The label is constant across all rows of H_i (the eventual disposition is a single hospitalization-level fact broadcast to every grid point).
- Because the Task T1 / Task T2 cohort filter guarantees $\texttt{disp}_i$ is either `"expired"`, `"hospice"`, or matches the LTACH regex, $\texttt{label_ltach}_i + \texttt{label_mortality}_i = 1$ for every retained row.

3 CLIF tables and categories used

The CLIF surface for Task T2 is identical to Task T1: the same three tables (`hospitalization`, `patient`, `adt`) and the same category-column filters. The cohort filter on `hospitalization.discharge_category`

keeps {"expired", "hospice"} ∪ regex ltach|long.?term (case-insensitive); the ADT filter restricts to location_category = "icu" (case-insensitive); patient is join-only. See the full table and per-table column inventory in the sibling document icu_daily_mortality.tex (§CLIF tables and categories used). Tasks T1 and T2 differ only in the final label predicate, not in which CLIF data they touch.

4 Algorithmic flow (shared with Task T1)

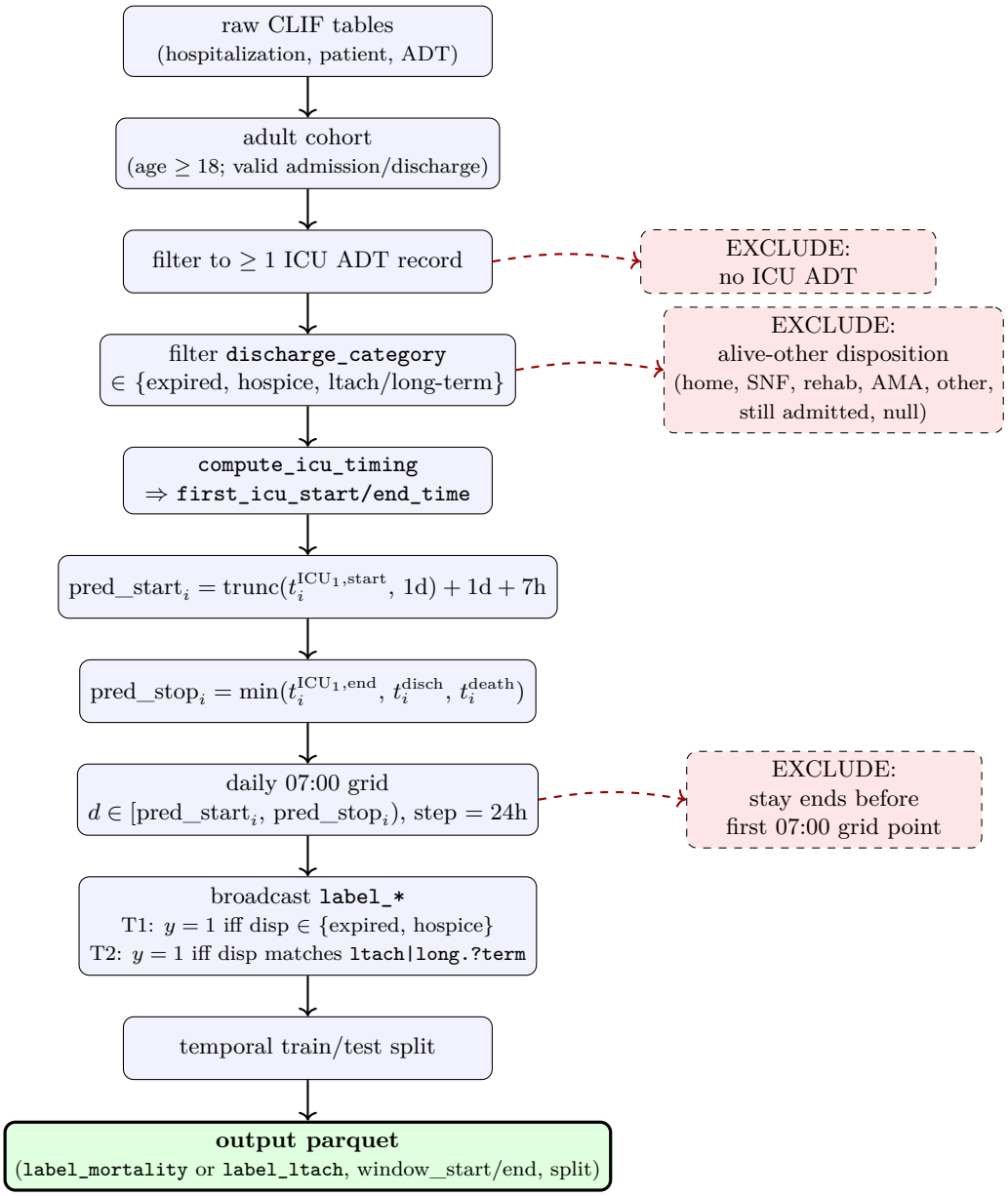


Figure 1: Build pipeline shared by Tasks T1 and T2. The label box at the bottom is parameterized: T2 emits $y = 1$ iff $\texttt{disp}_i$ matches `ltach|long.?term`. Everything else — cohort filters, ICU timing, pred_start / pred_stop, the daily 07:00 grid, the 24h lookback, the temporal split, the parquet schema — is identical to Task T1.

5 Design choices and rationale

Why a separate schematic for the complement label. A reader scanning the supplement should find one document per FLAIR task, with its own META, label rule, and prevalence numbers. The build pipeline being shared is a convenience of implementation, not a property of the task

definition: a future site whose `discharge_category` taxonomy splits LTACH from other long-term destinations more finely could change Task T2's regex without affecting Task T1.

Prevalence asymmetry vs. Task T1. At the row level the positive rate is 42.6% LTACH, complementary to T1's 57.4% mortality. At the stay level the asymmetry is sharper: 32.9% of retained stays are LTACH-positive vs. 67.1% mortality-positive (see `docs/paper/outline.tex` Table 1). In MIMIC, hospice and expired-in-hospital both fall on T1's positive side, while only the LTACH-or-long-term family falls on T2's, so the marginals are not 50/50. Reporting AUPRC alongside AUROC is mandatory.

Why LTACH is the operationally interesting label. LTACH transfer is a planned, resource-intensive disposition decision that triggers care-coordination work several days before discharge (Kahn et al., 2010). A morning prediction of LTACH-bound vs. in-hospital-mortality trajectory can drive social-work referrals, family conversations, and family-meeting timing; the same trajectory forecast from the mortality side (T1) is more commonly used for goals-of-care conversations. The two complementary framings expose the same underlying signal to two distinct downstream workflows.

6 Out of scope

- All items listed in `icu_daily_mortality.tex` (Task T1) §*Out of scope* carry over.
- Finer-grained LTACH-vs-skilled-long-term-care distinctions within the LTACH-positive class.

References

Jeremy M. Kahn, Nicole M. Benson, Dina Appleby, Shannon S. Carson, and Theodore J. Iwashyna. Long-term acute care hospital utilization after critical illness. *JAMA*, 303(22):2253–2259, 2010. doi: 10.1001/jama.2010.761.